A Criticality Claim That Does Not Survive Distribution Fitting: the Empedoclean Node Field Re-examined

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Abstract

An earlier internal report claimed that a deterministic node field with coupled attraction and repulsion exhibits self-organised criticality: a band of the repulsion parameter in which the cluster-size distribution follows a power law. That claim rested on a single seed, one node count and one time horizon. We re-run the sweep over 16 seeds and nine parameter settings and apply standard distribution fitting: discrete power-law maximum likelihood with the lower cutoff chosen by minimising the Kolmogorov–Smirnov distance, a lognormal fit to the same tail, and the Vuong log-likelihood-ratio test. The power law never wins. At three of the seven fittable settings the lognormal is significantly better; at the setting originally reported as critical the two are statistically indistinguishable, which is the absence of evidence for criticality rather than evidence for it. We further find that the regime label at the claimed critical band is not stable across seeds. The original positive result was a single-seed artifact of a heavy-tailed snapshot. We report the negative result, keep the falsified claim visible, and state what a criticality claim in a toy dynamical system would actually require.

1 What was claimed

The Empedoclean dyad is a deterministic dynamical system over a field of nodes with two forces: an attractive term ("Eros", strength `eros_strength`) and a repulsive term ("Nemesis", `nemesis_strength`). The original report described a 300-node, 2-dimensional, 400-tick sweep of the repulsion strength against a fixed attraction of 0.10, and reported three regimes: a supercritical regime in which one giant cluster forms, a subcritical regime in which the field fragments into singletons, and — between them — a *critical* band at nemesis ≈ 0.01 exhibiting a power-law cluster-size distribution. The reported evidence for that band was a single cluster-size snapshot, whose ten largest clusters were $59, 32, 9, 9, 8, 8, 7, 6, 6, 5$.

That evidence was always weak, and the original report said so: "a heavy-tailed snapshot under one configuration, not an SOC proof." The open questions it listed were multiple seeds, finite-size analysis, and distribution fitting with goodness-of-fit and confidence intervals. This paper answers them.

2 Method

2.1 Sweep

We use the same field, the same geometry, and the same link distance as the original: 300 nodes, 2 dimensions, 400 ticks, `eros_radius` 0.30, `eros_strength` 0.10, `nemesis_radius` 0.20, `max_step` 0.03, decay, death and synthesis disabled to isolate the attraction/repulsion balance. We sweep nemesis $\in \{0, 0.001, 0.003, 0.006, 0.01, 0.02, 0.04, 0.08, 0.16\}$ and run **16 seeds** per setting, yielding 144 independent runs.

2.2 Fitting

For each setting we pool cluster sizes across seeds and follow the procedure of Clauset, Shalizi and Newman (2009):

1. Choose the lower cutoff $x_{\min}$ by minimising the Kolmogorov–Smirnov distance between the empirical tail and the fitted discrete power law.
2. Fit the discrete power-law exponent α by maximum likelihood, normalising with the Hurwitz zeta function $\zeta(\alpha, x_{\min})$.
3. Fit a lognormal to the same tail by maximum likelihood.
4. Compare the two non-nested models with the Vuong log-likelihood-ratio test. A positive R favours the power law; a negative R favours the lognormal; a non-significant p means the two are indistinguishable on this data.

This is the test the original report named as the condition under which the criticality claim would be unfounded.

3 Results

nemesis	**clusters**	$x_{\min}$	α	**KS**	R	p	**verdict**
0.000	50	76	3.558	0.091	4.2	0.291	indistinguishable
0.006	81	2	1.201	0.534	−17.4	0.046	**lognormal better**
0.010	2507	82	4.872	0.166	−2.5	0.396	indistinguishable
0.020	4576	3	5.000	0.900	−0.9	0.856	indistinguishable
0.040	4722	2	5.000	0.987	−89.1	0.014	**lognormal better**
0.080	4747	2	5.000	0.918	−18.9	0.013	**lognormal better**
0.160	4758	2	5.000	1.000	—	—	undetermined

Table 1: Distribution fits, 16 seeds per setting. Settings 0.001 and 0.003 produce too few clusters to fit. R is the Vuong statistic; positive favours the power law.

The power law never wins. In no setting does R come out positive and significant. At three settings the lognormal is significantly better than the power law. At the setting originally reported as critical (nemesis = 0.010), the two models are indistinguishable ($R = -2.5$, $p = 0.40$) — which means this data provides no evidence distinguishing a power law from a lognormal, not that a power law holds.

Two further observations bear on the fits. First, α runs to the upper bound of the search in the fragmented regime and the KS distance approaches 1.0, indicating that a power law is simply the wrong family there. Second, $x_{\min}$ chosen at 2 or 3 in the fragmented settings means the fitted tail consists of almost all clusters, including the abundance of singletons — the classic failure mode where a heavy-tailed-looking distribution is fitted above a cutoff so low that the fit is meaningless.

3.1 The critical band is not stable across seeds

The original claim located a critical band at nemesis ≈ 0.01. Classifying each of the 16 runs at that setting with the code's own regime rule yields a mixture: some runs classify as *Critical* and some as

Subcritical. At no other setting does the classifier disagree across seeds. The mean largest-cluster fraction moves from 0.986 at nemesis = 0.006 to 0.346 at 0.010 to 0.009 at 0.020, so what the sweep actually shows is a *sharp* transition between two phases, not a self-organised band in which the system sits at a critical point. A sharp transition whose location varies with the seed is the opposite of the robustness that self-organised criticality is supposed to explain.

4 Why the original positive was an artifact

The original snapshot had ten clusters, of which the largest was 59 and the rest fell quickly. A snapshot of that shape is exactly what a lognormal with a heavy tail produces, and a handful of cluster sizes cannot distinguish the two families. With one seed there is also no way to see that the regime label is unstable, and with $n = 10$ clusters there is no power to fit a tail at all. The artifact was not a coding error: the measurement was correctly taken, and correctly flagged as insufficient. It became a claim only in the retelling.

5 What a criticality claim would require

We state this so the refutation has a constructive form. A defensible claim of criticality in a system like this needs, at minimum:

1. **Many seeds**, with the regime label stable across them at the claimed setting.
2. **Distribution fitting** against at least a lognormal alternative, with the cutoff chosen by a principled criterion rather than by eye, and goodness-of-fit reported.
3. **A finite-size analysis**: the cutoff behaviour of the cluster-size distribution should change with node count in a specified way if the system is critical.
4. **An order parameter and its scaling**, not merely a heavy-tailed snapshot.
5. **Control systems** whose exact critical point is known, run through the same analysis pipeline, to show the pipeline can detect criticality when it is present.

Point 5 is the one we regard as most valuable and least often done: without a positive control, a null result cannot be distinguished from an insensitive instrument. We have not run it, so we cannot claim our pipeline would have detected a genuine power law; we claim only that it does not detect one here.

6 What survives

The system is not worthless because the criticality claim failed. What survives is narrower and better supported:

- The attraction/repulsion balance produces a sharp transition between a collapsed phase and a fragmented phase, stable across 16 seeds at every setting except the transition region.
- The transition region is seed-sensitive, which is a property worth knowing before anyone builds on the system at that setting.

- The promotion gate the project had already imposed on this module (gate 2, no-regression, still unpassed) was the right call; the module remains research-only and touches no production path.

7 Reproducibility

- Sweep: `examples/empedocles_multiseed.rs`, run as `cargo run --release --features empedocles --example empedocles_multiseed -- 16`, writing `research/empedocles_multiseed.json`.
- Fits: `research/papers/fit_empedocles.py`, writing `research/empedocles_fits.json`.
- The example is declared in `Cargo.toml` with `required-features = ["empedocles"]`, matching the two pre-existing Empedocles examples.
- The original single-seed example, `examples/empedocles_criticality.rs`, is unchanged and still builds. Its output is the artifact this paper refutes; we have kept it rather than deleted it.

8 Conclusion

A single-seed heavy-tailed snapshot was reported as evidence of self-organised criticality in a 300-node dynamical system. Re-running the sweep with 16 seeds and fitting the resulting distributions properly shows that the power law never beats a lognormal, that at the claimed critical setting the two are indistinguishable, and that the regime label at that setting is unstable across seeds. The claim is refuted. The system's interesting behaviour — a sharp, seed-influenced transition between collapsed and fragmented phases — survives, but it is a phase transition, not self-organised criticality. We keep the refuted claim, the original example and the original report in the repository, because the sequence from plausible snapshot to refuted claim is the most useful thing this line of work produced.

References

[Clauset et al.(2009)] Clauset, A., Shalizi, C. R., & Newman, M. E. J. (2009). Power-law distributions in empirical data. *SIAM Review*, 51(4), 661–703. DOI: 10.1137/070710111.